



GLOBALPAY

Mandatory Architecture & Engineering Deliverables

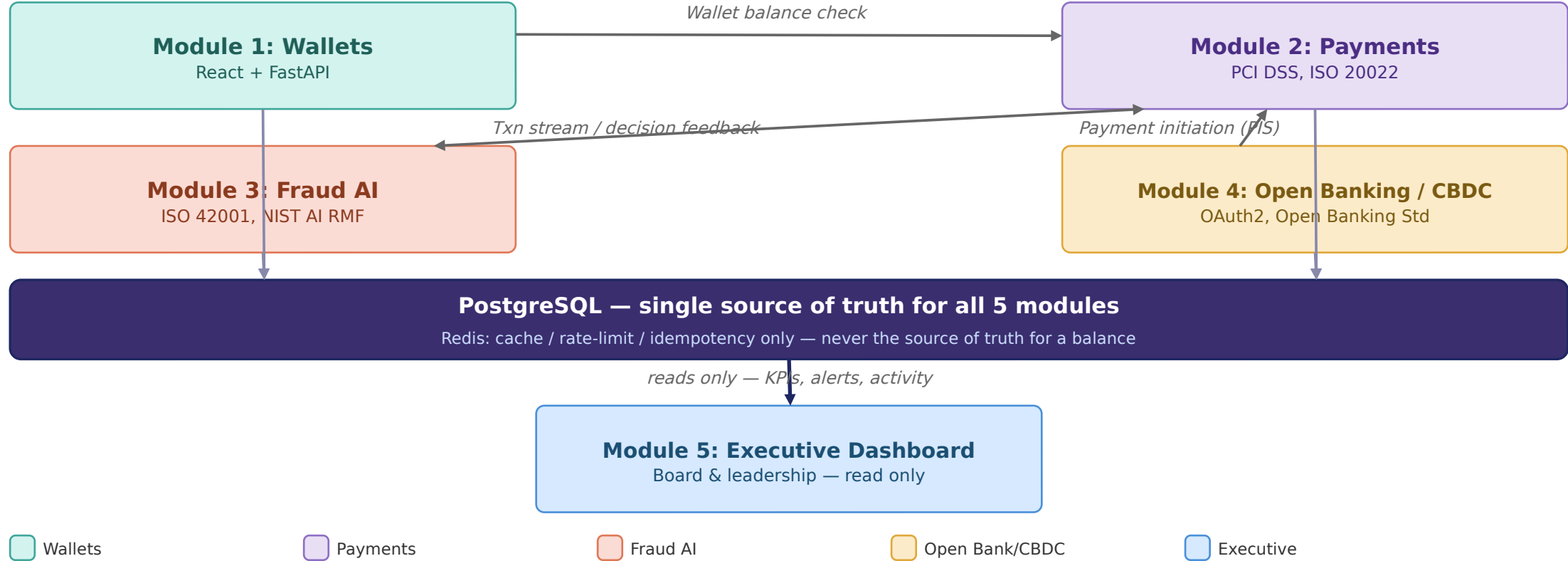
All 17 mandatory diagrams, numbered exactly as listed in the exam brief

#	Diagram	Also in Stage 1 deck?
1	Enterprise Financial Technology Architecture	Yes
2	High-Level Solution Architecture	No - here only
3	Detailed System Architecture	No - here only (composite index)
4	Digital Wallet Architecture	Yes (Module 1)
5	Payment Processing Architecture	Yes (Module 2)
6	Open Banking API Architecture	Yes (Module 4)
7	AI Fraud Detection Architecture	Yes (Module 3)
8	CBDC Simulation Architecture	No - here only
9	Enterprise Network Architecture	Yes
10	Network Flow Diagram	Yes (combined in Module 11)
11	Data Flow Diagram (Level 0)	Yes (combined)
12	Data Flow Diagram (Level 1)	Yes (combined)
13	Digital Payment Workflow	Yes (combined in Module 2)
14	Fraud Detection Workflow	Yes (combined in Module 3)
15	Open Banking API Workflow	Yes (combined in Module 4)
16	Database ERD	Yes
17	Sequence Diagram	No - here only

G Enterprise Financial Technology Architecture

1

How the five modules connect

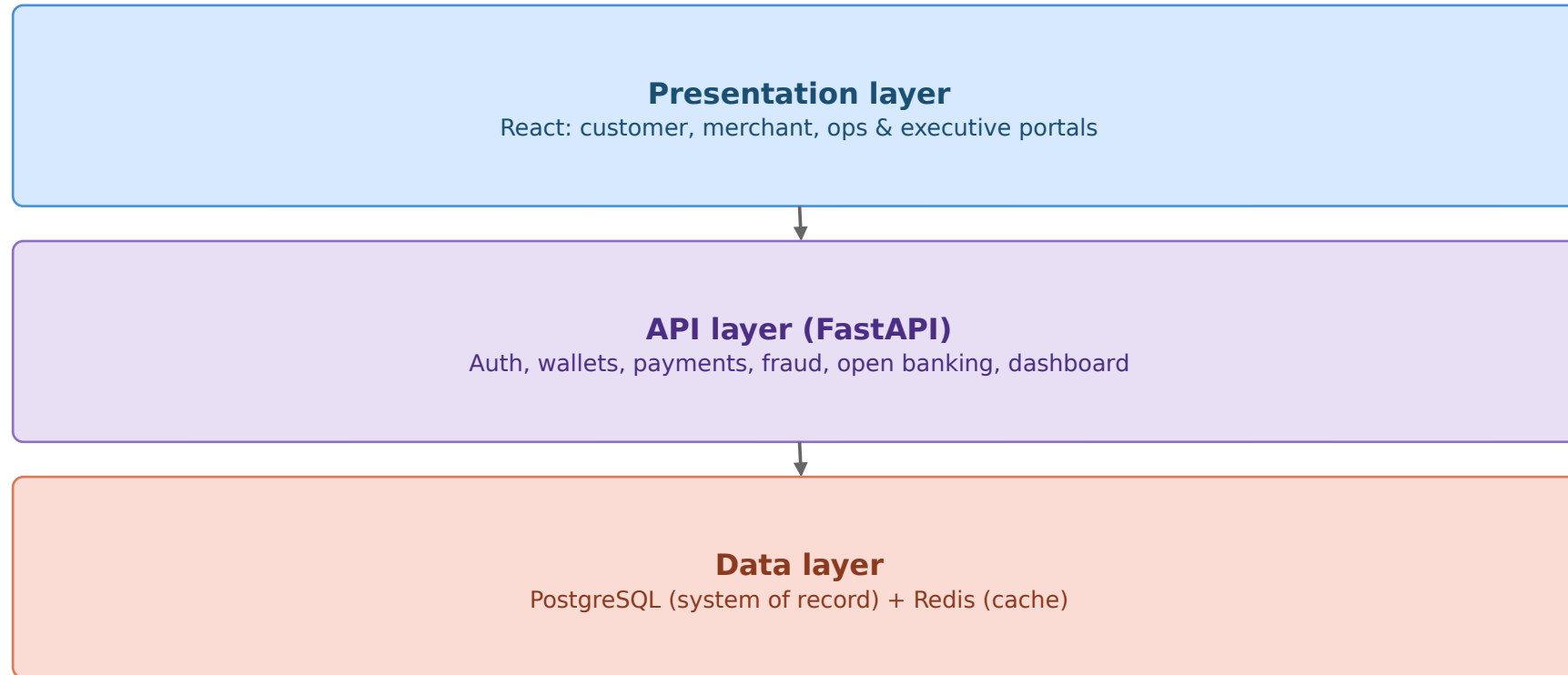


Modules 1-4 write to PostgreSQL; Module 5 reads only; only the named service calls above cross module boundaries

G High-Level Solution Architecture

The 30,000-ft view

2



Security & compliance

ISO 27001 / 27017 / 27018
PCI DSS v4.0*
OWASP Top 10 + API Top 10
ISO 42001 / NIST AI RMF
NIST CSF 2.0 + CIS Controls
COBIT 2019

This is the 30,000-ft view — each module diagram (4-7) zooms into one API layer service

G Detailed System Architecture

3

Composite reference — each module is fully detailed in diagrams 4-8

Module 1

Wallets

React UI -> FastAPI (auth, customers, wallets, merchants) -> PostgreSQL

Module 2

Payments

Validation, transaction engine, double-entry ledger -> ledger tables

Module 3

Fraud AI

Feature engineering -> Logistic Regression -> risk scoring -> human review

Module 4

Open Banking/CBDC

AIS/PIS APIs, CBDC API, activity monitoring -> separate CBDC ledger

Module 5

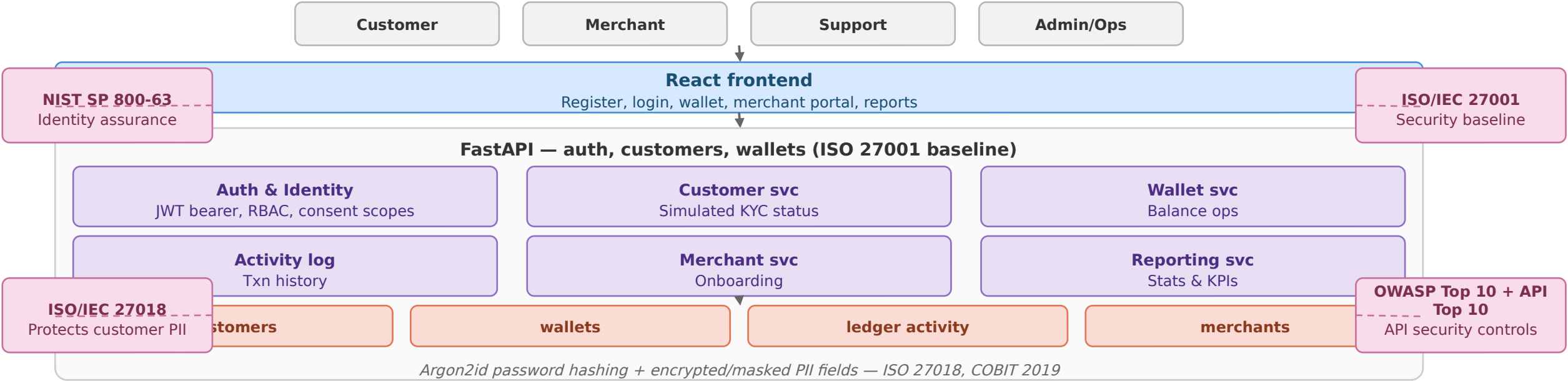
Executive Dashboard

Read-only aggregator across Modules 1-4 -> executive reports

This slide exists to index the detail, not duplicate it — see diagrams 4-8 for full architecture, framework badges, and data model per module

G Digital Wallet Architecture

Registration, wallet creation, balance operations, merchant accounts

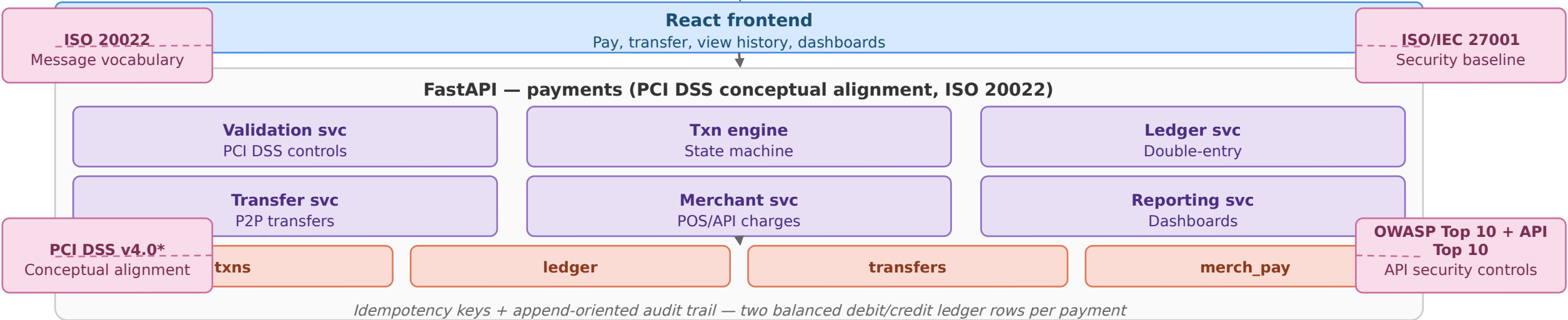


Writes directly into PostgreSQL — see diagram 1 for the full enterprise data-flow context

G Payment Processing Architecture

5

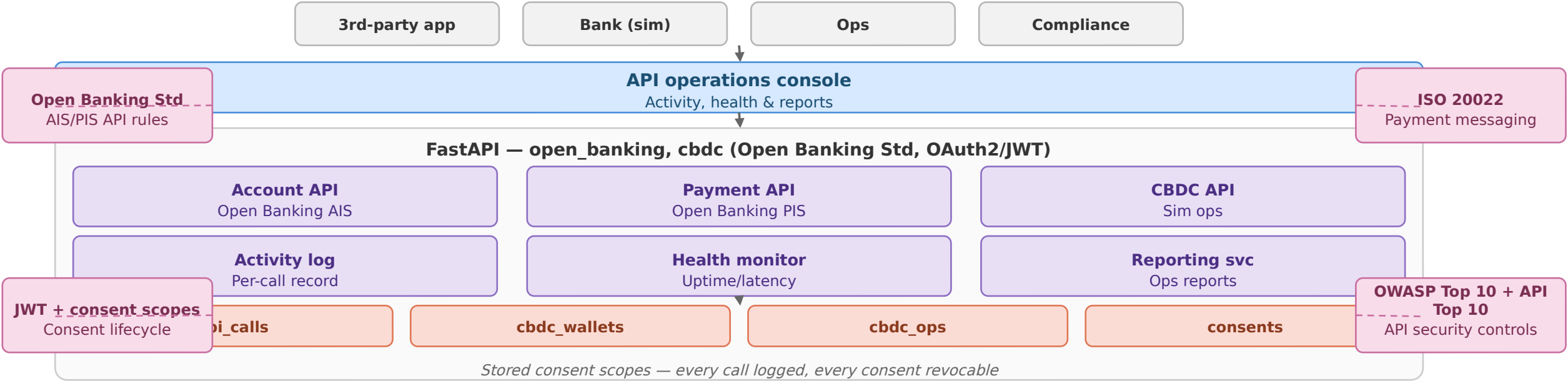
Validation, double-entry ledger, transfers, merchant charges



Direct link to Module 3 for fraud scoring — the only true module-to-module call besides Module 4's PIS

G Open Banking API Architecture

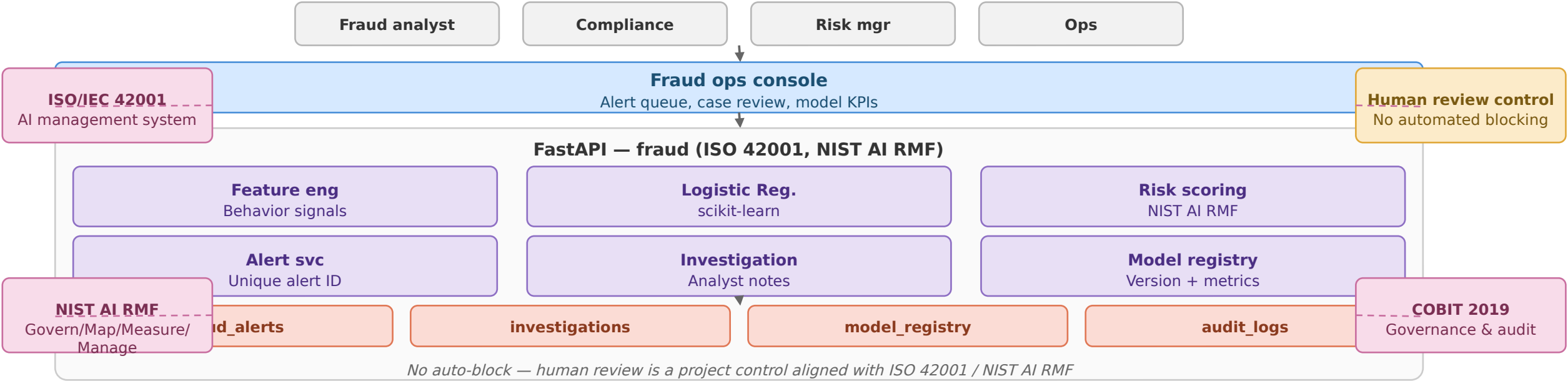
Account info & payment initiation APIs, CBDC wallet ops



PIS calls Module 2's transfer logic directly rather than duplicating it — see diagram 8 for CBDC's separate ledger

G AI Fraud Detection Architecture

Feature engineering, classification, mandatory human review



Surfaces to fraud/risk/compliance teams only — never shown to customers

G CBDC Simulation Architecture

8

Kept separate from the commercial wallet ledger

OAuth2/JWT, ISO 20022
Access + messaging

COBIT 2019
Governance & audit



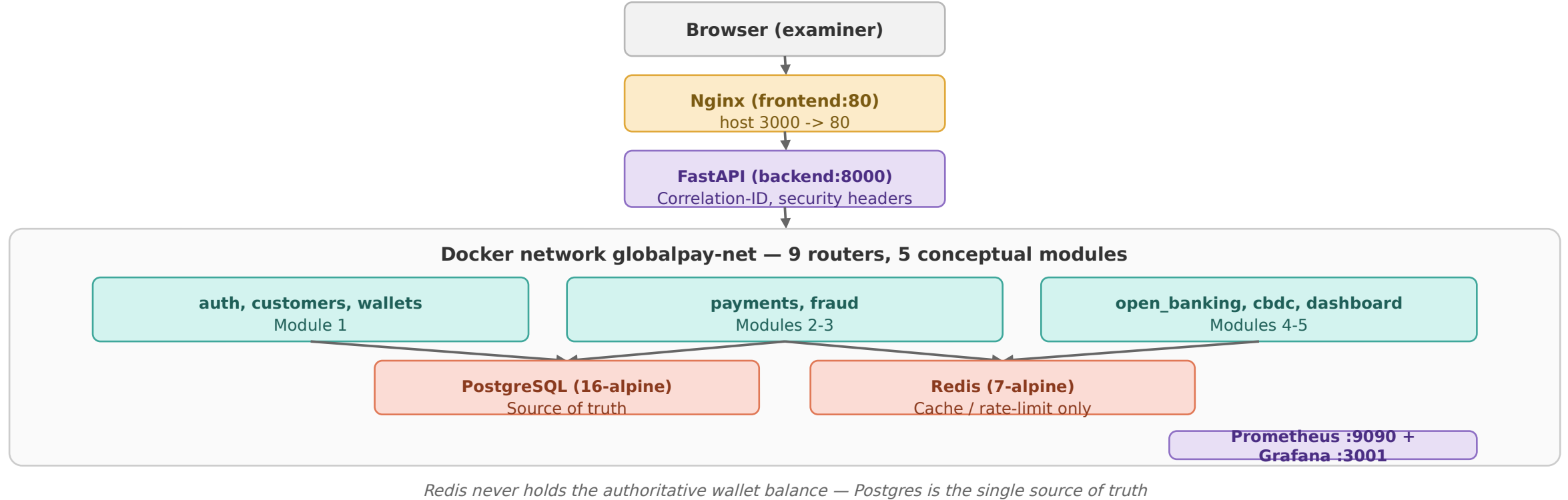
Reporting reads `cbdc_wallets` and `cbdc_operations`; these records remain separate from commercial wallet data

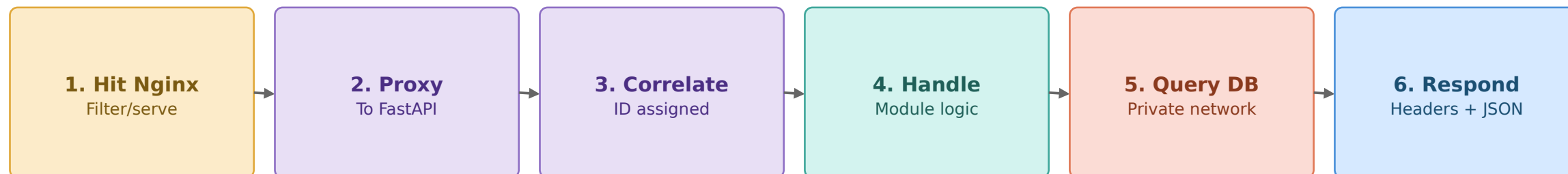
Central-bank money is never mixed with commercial wallet balances — this table separation is the audit boundary

G Enterprise Network Architecture

9

compose.yaml — 6 containers on one Docker network (globalpay-net)

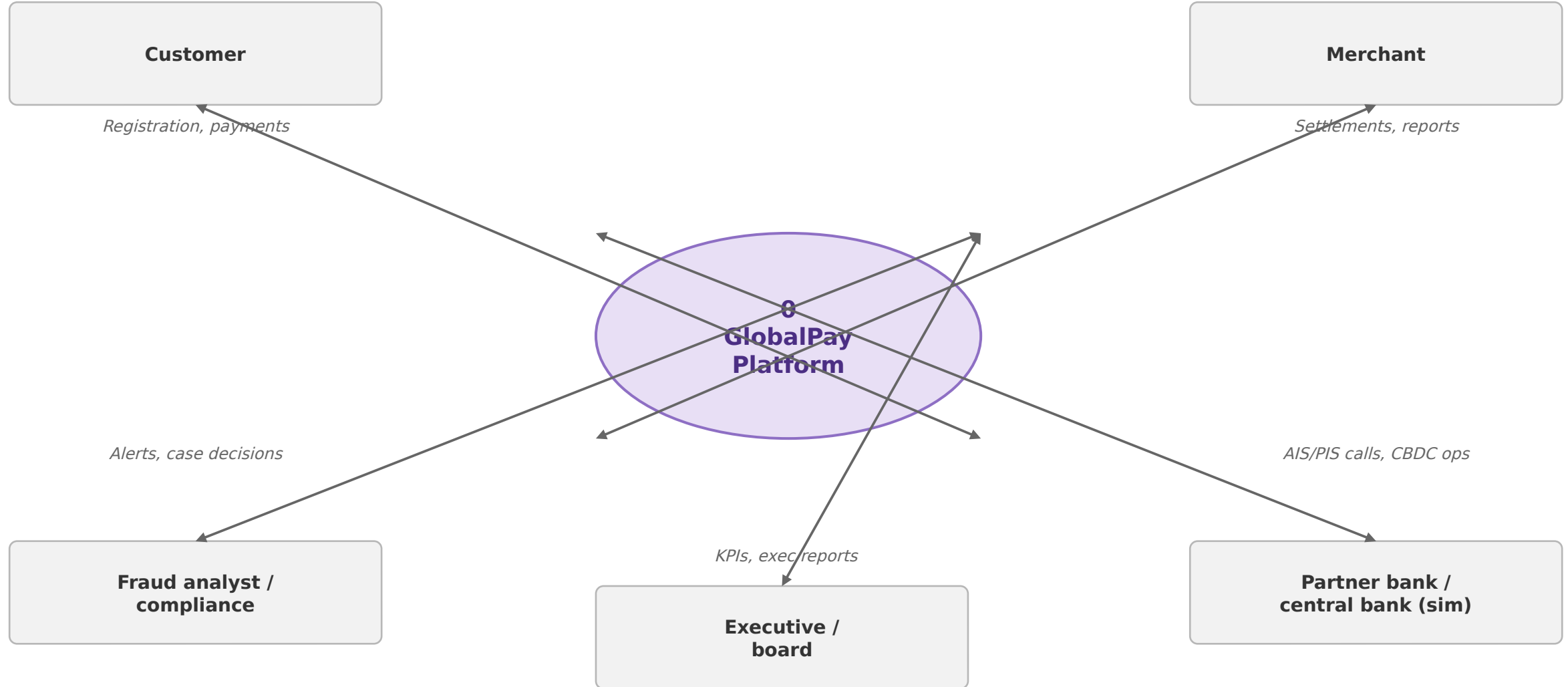




G Data Flow Diagram — Level 0 (Context)

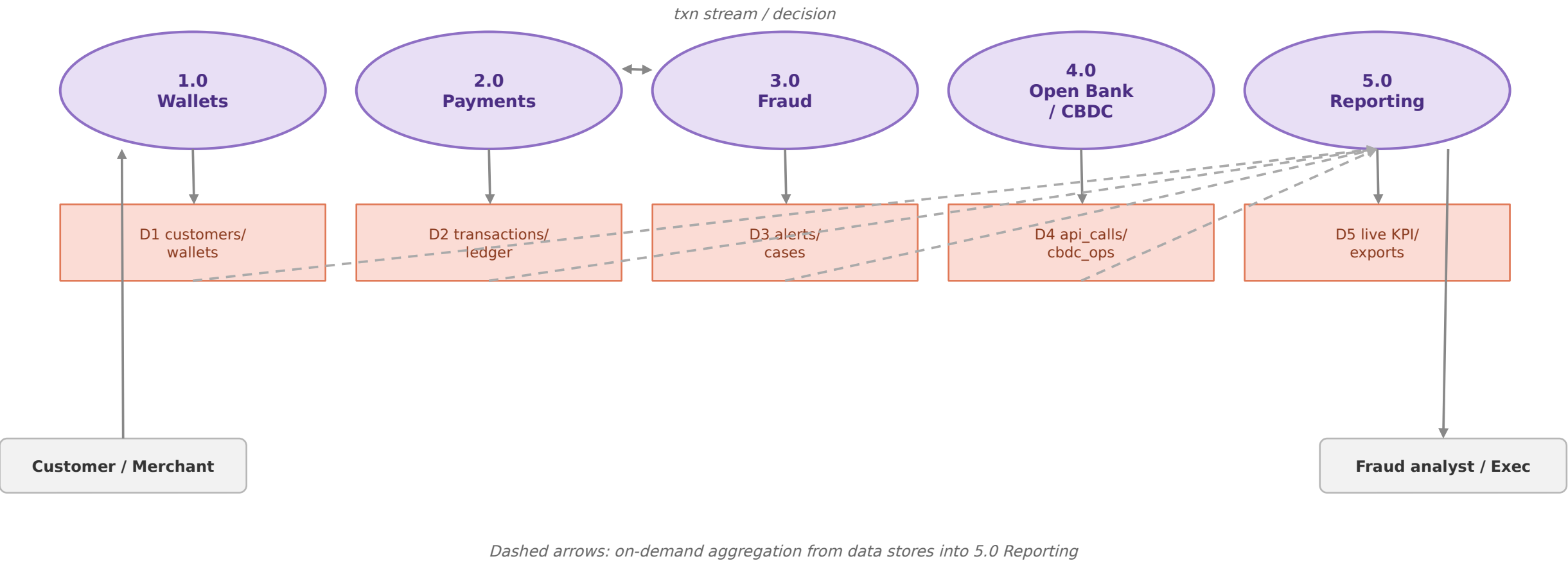
11

The platform as a single process, with its external entities



G Data Flow Diagram — Level 1

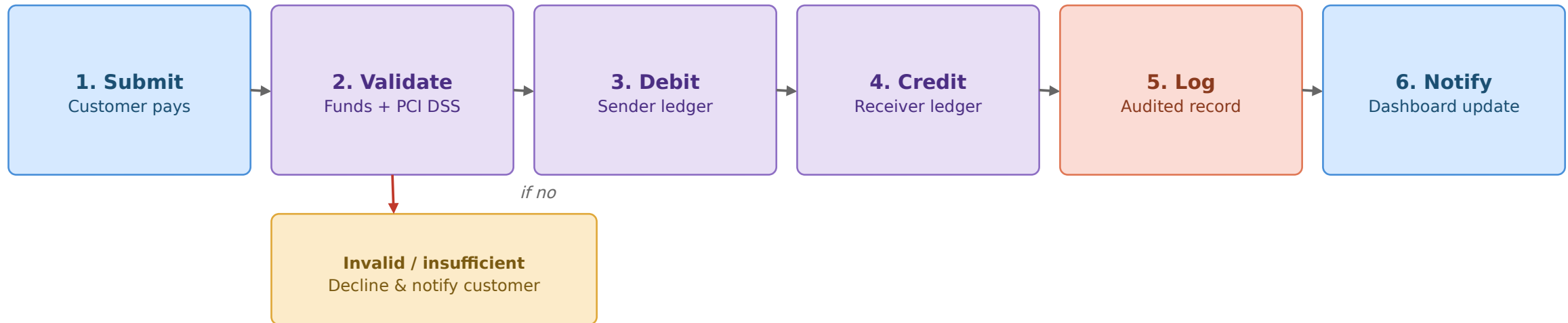
The platform decomposed into its five core processes

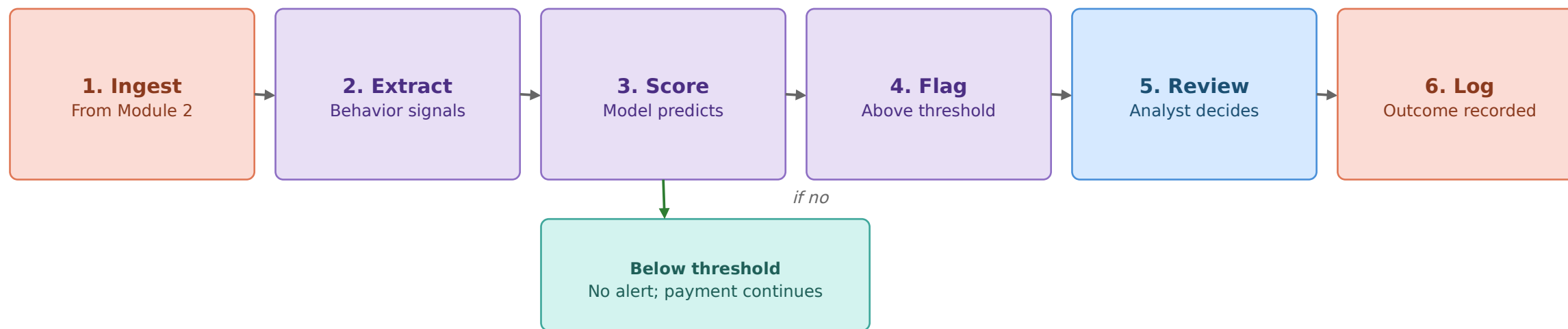


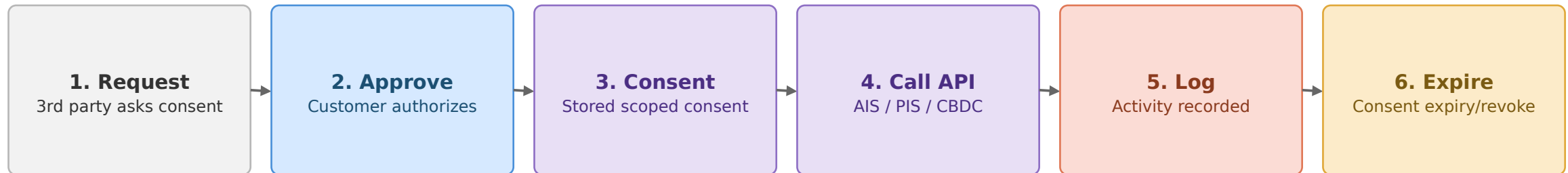
G Digital Payment Workflow

Validation branch shown explicitly

13



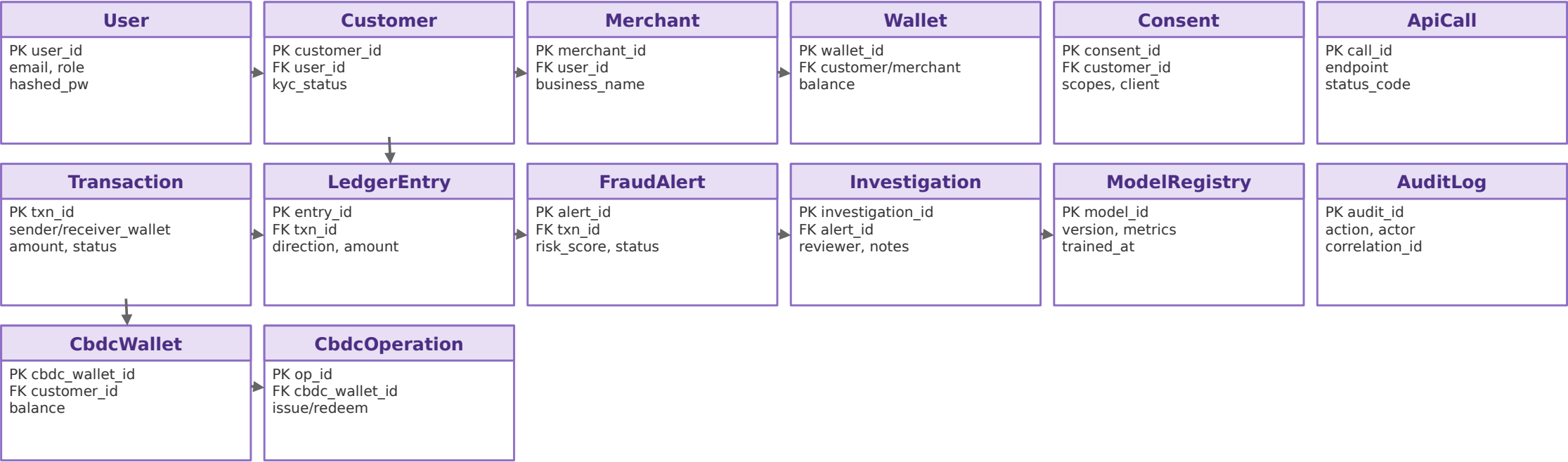




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Database Entity Relationship Diagram (ERD)

14 tables from backend/app/models/entities.py



User -> {Customer, Merchant}; Customer -> {Wallet, Consent, CbdcWallet}; Merchant -> Wallet. Transaction -> {LedgerEntry, FraudAlert -> Investigation}. ModelRegistry and AuditLog are cross-cutting.

End-to-end lifecycle across all five modules

